

## Problem 2(b)/(c): Changing and breaking scaling laws

30 new runs (10 for (b), 20 for (c)) + 2(a) baselines. Model-size axis: d4-d9 at 614M tokens. Data axis: d6 (16M params) with 3M-614M tokens. Slopes = exponent of a straight-line fit on log-log axes; 'bend' = mid-point residual (0 = perfectly straight).

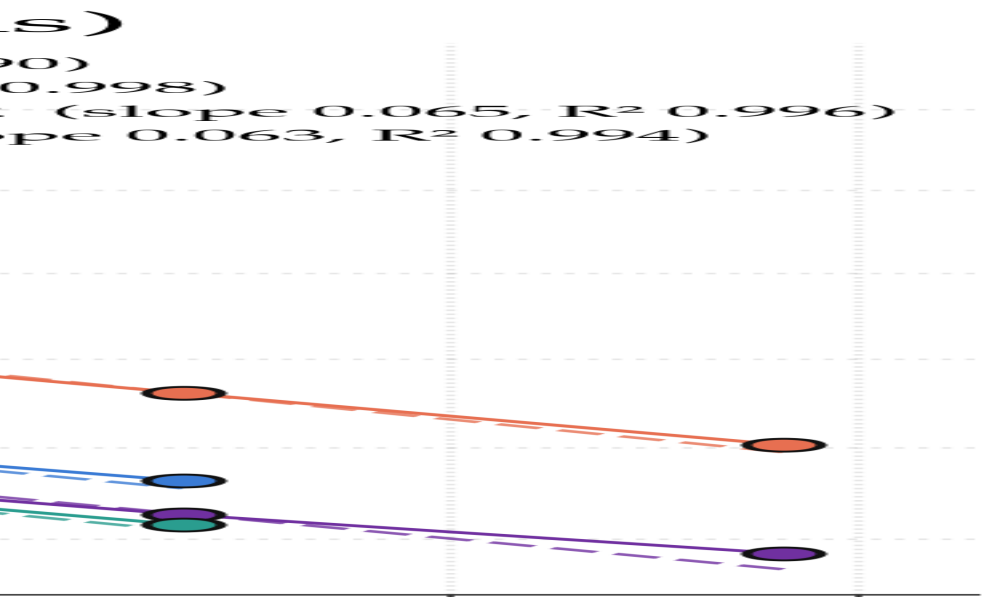
### Model-size slopes on matched depths (d4, d6, d8)

| Intervention                               | d4    | d6    | d8    | Slope  | Bend    | Verdict                      | Mechanism                                           |
|--------------------------------------------|-------|-------|-------|--------|---------|------------------------------|-----------------------------------------------------|
| Baseline (2a)                              | 3.290 | 3.057 | 2.926 | 0.067  | -0.005  | reference                    | -                                                   |
| Dropout 0.2 (2a)                           | 3.452 | 3.216 | 3.061 | 0.068  | -0.002  | offset only                  | Capacity tax; slightly shrinking with size          |
| Tuned: $\beta_2$ .99, warmup 4%, wd .2 (b) | 3.270 | 3.047 | 2.915 | 0.065  | -0.004  | offset only                  | Better optimizer helps every size about equally     |
| bs 256, LR 6e-3, warmup 4% (b)             | 3.313 | 3.088 | 2.963 | 0.064  | -0.005  | offset only                  | Fewer steps cost every size about equally           |
| Weight decay 2.0 (c)                       | 3.355 | 3.148 | 3.039 | 0.056  | -0.005  | <b>slope change, linear</b>  | Shrink rate $lr \times wd$ limits large models more |
| LR .03, no warmup, no clip (c)             | 3.317 | 3.133 | 3.012 | 0.055  | -0.002  | <b>slope change, linear</b>  | Early instability costs more for wider models       |
| Constant LR (2a)                           | 3.421 | 3.243 | 3.172 | 0.043  | -0.007  | <b>slope change, ~linear</b> | No annealing; noise floor grows with size           |
| LR 0.03 (2a)                               | 3.321 | 3.105 | 3.075 | 0.045  | -0.016  | slope change, bending        | LR too high for wide models                         |
| LR 0.1 (c)                                 | 3.407 | 3.241 | 3.144 | 0.046  | d9 = d8 | <b>breaks at d9</b>          | Saturates: d8 3.144, d9 3.146                       |
| 37.5k seqs $\times$ 16 epochs (c)          | 3.393 | 3.367 | 3.513 | -0.018 | -0.018  | <b>breaks: U-shape</b>       | Memorization: best at d6, d9 = 3.640                |
| bs 1024, 586 steps (c)                     | 3.767 | 4.004 | 4.293 | -0.073 | -0.008  | <b>breaks: inverse</b>       | Step-starved: bigger = worse                        |

### Data-axis slopes (d6)

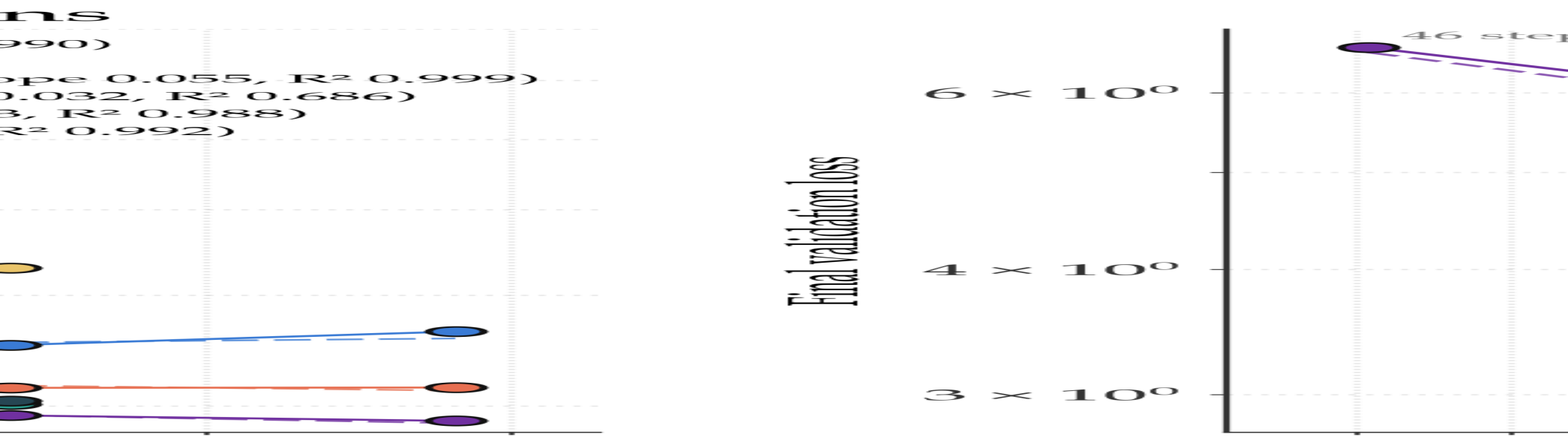
| Series                  | Tokens (steps)                          | Loss                  | Slope | Verdict                                                      |
|-------------------------|-----------------------------------------|-----------------------|-------|--------------------------------------------------------------|
| Baseline, 154M-614M     | 154M / 307M / 614M (2.3k / 4.7k / 9.4k) | 3.329 / 3.167 / 3.057 | 0.061 | linear ( $R^2$ 0.990)                                        |
| Dropout 0.2, 154M-614M  | same                                    | 3.501 / 3.333 / 3.216 | 0.061 | offset only: identical data slope                            |
| Tiny data, 3M-31M       | 3M / 10M / 31M (46 / 156 / 468 steps)   | 6.660 / 5.599 / 4.486 | 0.17  | <b>breaks: ~3x steeper below ~2k steps</b>                   |
| 16x repeated 38M unique | 614M seen (9.4k steps)                  | 3.367                 | -     | beats 31M single-epoch (4.486), loses to 614M unique (3.057) |

2(b): Slope changes inside the linear regime



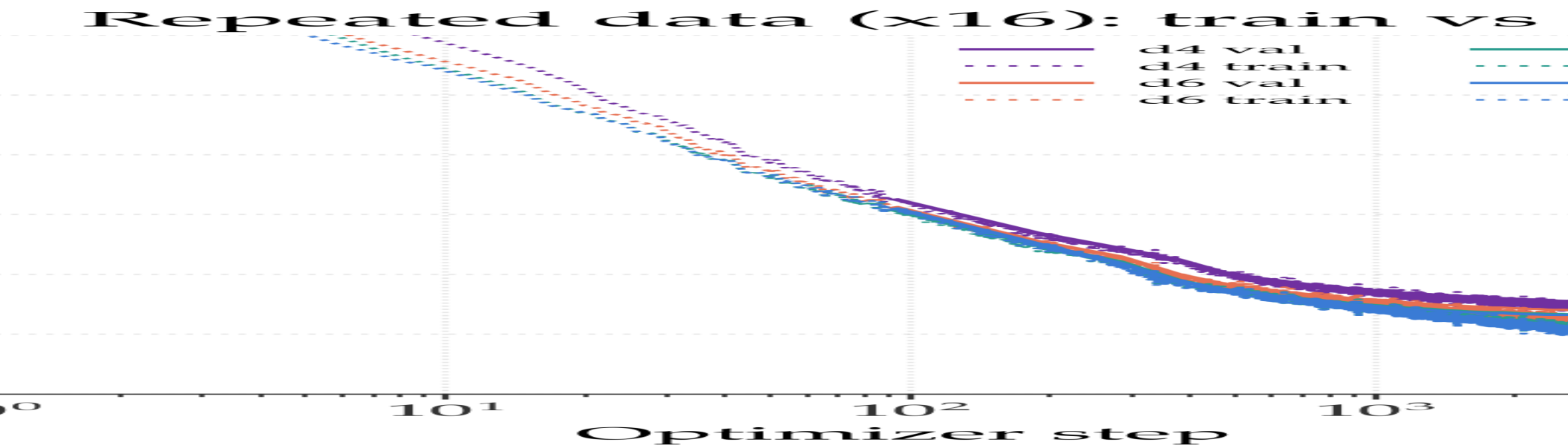
Left: the three planned (b) interventions (tuned, bs 256) and dropout shift the curve but keep the slope (0.064-0.068 vs 0.067 on matched depths). Right: on the data axis dropout is also a pure offset (both slopes 0.061). The interventions that **change the slope while staying straight** are ones whose cost grows with model size: weight decay 2.0 (0.056), unstable LR 0.03 (0.055), constant LR (0.043) - see table.

2(c): Breaking the power law



Left: repeated data turns into a U (best at d6), bs 1024 inverts the trend (bigger is worse), LR 0.1 saturates (d8 = d9). Right: on the data axis the power law holds from 154M to 614M tokens, but below ~2k optimizer steps the curve steepens sharply - tiny budgets are in a different, optimization-limited regime. The diamond is 38M unique tokens repeated 16x.

## 2(c): Why they break - training dynamics



Left: at LR 0.1 all sizes follow nearly the same trajectory; the extra capacity of d8/d9 is never used because the step size, not the model, limits progress. Middle: with 16 epochs, train loss (dotted) keeps falling while val loss turns up at the end for d8/d9 - classic overfitting, and bigger models memorize more. Right: at bs 1024 every model sits on an early plateau near loss 6.9 before learning takes off, and larger models stay there longer; with only 586 steps they never catch up.